

INTRODUCTION

Myelin oligodendrocyte glycoprotein antibody-associated disease (MOGAD) is a rare, immune-mediated central nervous system (CNS) disorder characterized by recurrent episodes of demyelination affecting the optic nerves, brain, and spinal cord. Initially recognized as a distinct entity in the past two decades, MOGAD accounts for a subset of non-multiple sclerosis demyelinating diseases and is distinct from neuromyelitis optica spectrum disorder (NMOSD). It is defined by the presence of autoantibodies targeting myelin oligodendrocyte glycoprotein (MOG), a protein expressed on the surface of myelin sheaths. Clinical manifestations of MOGAD are diverse and can include optic neuritis, transverse myelitis, and acute disseminated encephalomyelitis (ADEM). The diagnosis of MOGAD relies on the identification of MOG-specific antibodies using highly sensitive cell-based assays. Importantly, the absence of aquaporin-4 (AQP4) antibodies, which are diagnostic of NMOSD, further supports the diagnosis. Treatment strategies for MOGAD focus on managing acute relapses with high-dose corticosteroids and preventing recurrence through long-term immunosuppressive therapies. Unlike multiple sclerosis, disease-modifying therapies used in MS are ineffective and may worsen MOGAD.

CASE PRESENTATION

We present the case of a 38-year-old male who initially presented to the ED with aphasia, behavioral changes, and right hand numbness. A contrast-enhanced MRI of the brain revealed multiple white matter lesions, predominantly in the left hemisphere and more prominent in the left temporal lobe. Many of these lesions showed enhancement, consistent with acute demyelination. A lumbar puncture was performed, which revealed an elevated protein level (109 mg/dL) but with normal white blood cell count (3 cells/ μ L) and glucose (63 mg/dL). The CSF was negative for infectious organisms, and two oligoclonal bands (OCBs) were found, exclusive to the CSF, which is a hallmark finding in demyelinating diseases. The patient was initially empirically treated for meningitis with IV antibiotics but was transitioned to intravenous methylprednisolone (IVMP) after a negative meningitis panel. Post-treatment, the patient showed improvement but did not return to baseline. He continued to experience subtle word-finding difficulties, dysarthria, and mild imbalance, though he denied any focal weakness. A follow-up MRI confirmed the resolution of previous lesions, but new areas of hyperintense lesions on FLAIR imaging were noted, indicating ongoing disease activity. Throughout this period, the patient had no history of febrile illness or vaccinations prior to symptom onset, and his family history was non-contributory.

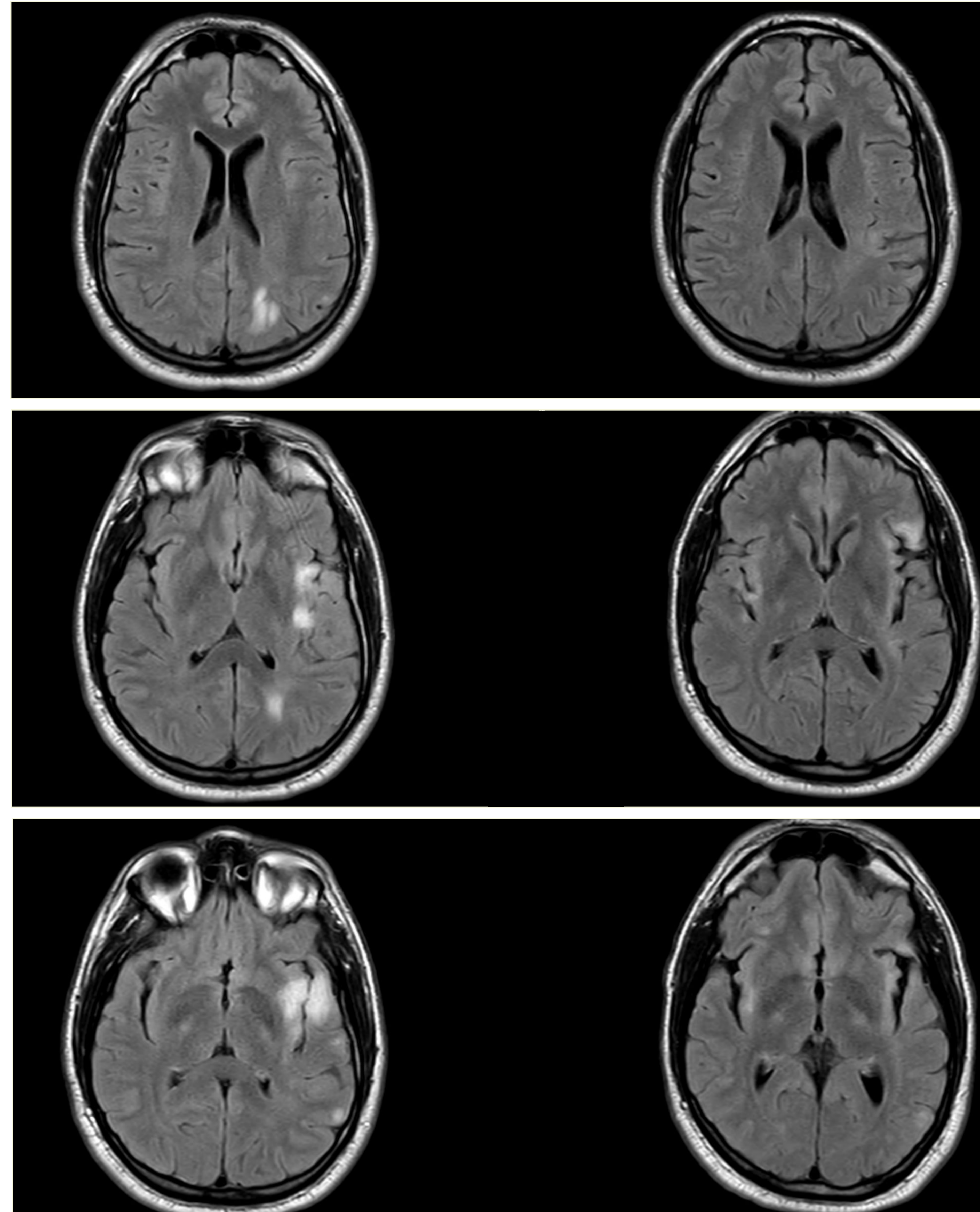


Figure 1 Axial FLAIR brain MRI images, taken 18 months apart, illustrate the relapsing-remitting nature of MOGAD, the images on the left are from 18 months prior, showing multifocal hyperintense lesions involving the cortical and subcortical white matter, with enhancement indicative of active demyelination, the images on the right, taken 18 months later, demonstrate resolution of prior lesions and the emergence of new hyperintense lesions

CLINICAL PEARLS

- Unlike MS, which predominantly affects periventricular white matter, **MOGAD often involves cortical and subcortical regions**
- In multiple sclerosis (MS), lesions are often round or ovoid in shape and uniformly distributed, in contrast, **MOGAD lesions tend to be larger and more irregular**
- Patients with MOGAD are **highly sensitive to corticosteroid tapering**, and relapses often occur if the taper is too rapid or steroids are discontinued prematurely
- **MS therapies (e.g., interferons, glatiramer acetate, natalizumab, fingolimod) are not effective in MOGAD**, MS therapies target T-cell and B-cell-mediated pathways involved in MS-specific inflammation but do not address the antibody-mediated pathology characteristic of MOGAD
- Unlike MS, which often has a gradual and subclinical onset, MOGAD typically presents with **acute, severe, and highly symptomatic episodes**, often mimicking encephalitis or transverse myelitis
- While the presence of MOG antibodies on cell-based assays (CBA) is diagnostic, up to **10-20% of patients with clinical and radiological features of MOGAD may be seronegative**
- Regular follow-up with serial MRI and clinical monitoring is essential in MOGAD especially in seronegative cases where antibody titers cannot be tracked

DISCUSSION

Given the clinical presentation, MRI findings, and relapsing-remitting course, the differential diagnosis included seronegative MOGAD versus atypical multiple sclerosis (MS). Notably, repeated testing for MOG antibodies via cell-based assays (CBA) was negative, but the MRI findings remained highly suggestive of MOGAD, particularly the cortical involvement and enhancement. Additional autoimmune and infectious workup, including ANA, anti-dsDNA, AQP4 antibodies, and a broad infectious panel, all returned negative results. Despite the absence of positive MOG antibodies, the patient's clinical course with relapses, the pattern of lesions, and the response to steroids strongly pointed toward a diagnosis of seronegative MOGAD.

REFERENCES

Please scan the adjacent QR code to view the references.

